Literature Status: A Strong Spacetime Variant of the Deift Conjecture Is False

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Status: literature-implied answer (partial variant). The separate later paper arXiv:2209.07501 explicitly cites the counterexample program arXiv:2111.09345 and disproves the strong spacetime/spatial-preservation form of the Deift conjecture. It does not fully settle the source's global temporal almost-periodicity statement.

1 Source conjecture and program

Damanik, Lukić, Volberg, and Yuditskii, *The Deift Conjecture: A Program to Construct a Counterexample*, arXiv:2111.09345v1 [1], state on arXiv PDF page 2:

Source Conjecture 1.1. *For the KdV equation*

$$\partial_t u - 6u\partial_x u + \partial_x^3 u = 0$$

with almost-periodic initial data $u(x, 0) = V(x)$*, the solution evolves almost periodically in time.*

The paper does not claim a counterexample. It proposes a coordinated program involving linearization beyond the direct-Cauchy-theorem regime, construction of a special Widom domain, stability of time frequencies, and an explicit description of singular characters. The last description is identified in the source as the main obstacle.

2 Separate later result

Chapouto, Killip, and Vişan, *Bounded solutions of KdV: uniqueness and the loss of almost periodicity*, arXiv:2209.07501v1 [2], explicitly cite arXiv:2111.09345 as the preceding program and say that they take a different, simpler route. They formulate the strong prediction that the KdV solution is almost periodic in spacetime, which in particular would make every spatial profile and every temporal trace almost periodic. On arXiv PDF page 7 they state:

Later Theorem 1.3. *There is a bounded solution* $q : [-T, T] \times \mathbb{R} \to \mathbb{R}$ *of KdV with almost-periodic initial data for which* $x \mapsto q(t_0, x)$ *is not almost periodic at some* $t_0 \in [-T, T]$.

The authors immediately identify this as a negative answer to the strong spacetime formulation. Their unconditional uniqueness theorem also rules out replacing this bounded solution by a better-behaved development of the same initial datum. Thus the later theorem gives a genuine KdV counterexample to spatial preservation, and hence to joint spacetime almost periodicity, while bypassing the source's Widom/DCT program.

3 Why the scope is partial

The exact temporal statement in arXiv:2111.09345 is stronger in a different direction: Bohr almost periodicity in t is a global property on $\mathbb{R}$. Theorem 1.3 of the later paper constructs its KdV solution only on $[-T, T]$. On arXiv PDF page 8, the authors explicitly state that they do not know whether this solution exists globally in time. They do obtain a temporal non-almost-periodicity statement for the corresponding *Airy* evolution, but not a global KdV temporal trace.

Consequently, spatial loss at one time disproves the strong spacetime formulation but does not, by itself, prove that a globally defined KdV trace $t \mapsto q(t, x)$ fails to be almost periodic. Nor does the later paper solve the source's specific Widom-domain Problem 1.6 or its auxiliary conjectures. This packet therefore records a partial-variant literature answer, not a full resolution of Conjecture 1.1.

4 Search evidence

The four lightweight run indexes contained no prior treatment of either arXiv id. A bounded exact-title, author, Deift-conjecture, and KdV-counterexample search through 12 August 2026 found arXiv:2209.07501 as the decisive later paper. Its citation of arXiv:2111.09345 and its own scope language make the relationship explicit; the remaining temporal limitation follows from its local time interval and global-existence disclaimer. No new mathematical theorem is claimed here.

References

[1] D. Damanik, M. Lukić, A. Volberg, and P. Yuditskii, *The Deift Conjecture: A Program to Construct a Counterexample*, arXiv:2111.09345v1 (2021).

[2] A. Chapouto, R. Killip, and M. Vişan, *Bounded solutions of KdV: uniqueness and the loss of almost periodicity*, arXiv:2209.07501v1 (2022).